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(54) **CONFIGURABLE DISPLAY HOLDING
APPARATUS AND SYSTEM**

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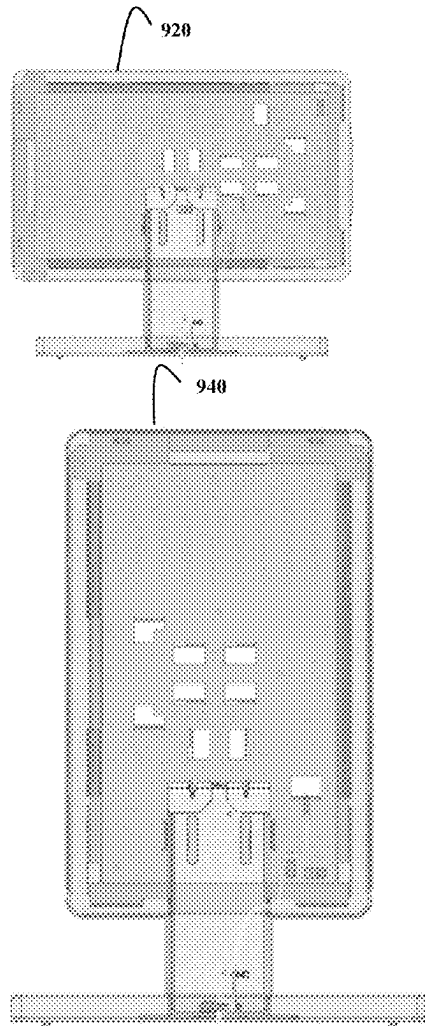
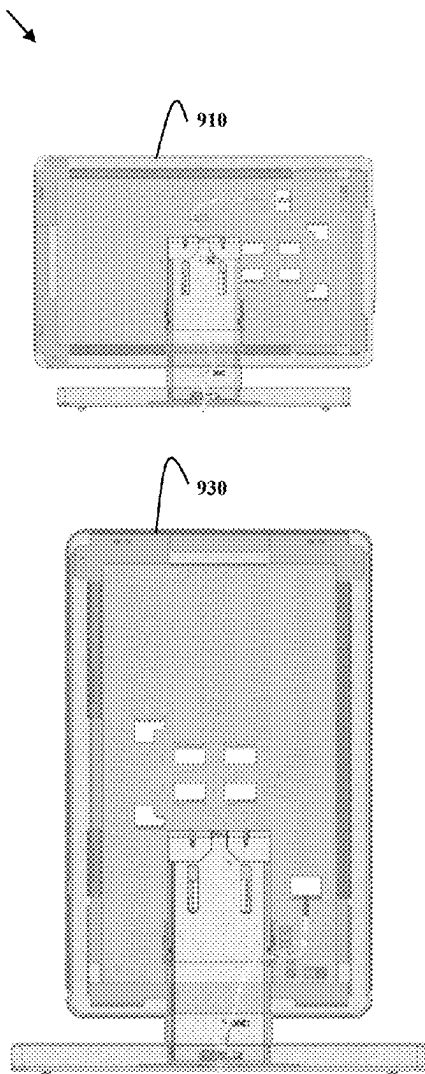
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(57)

ABSTRACT

A display holding apparatus includes a base, a vertical member, and a bracket. The vertical member is attachable to the base in one of two different orientations to form a stand. The bracket is attachable to the vertical member of the stand in one of four different orientations. The four different bracket orientations includes a low landscape orientation, a high landscape orientation, a low portrait orientation, and a high portrait orientation. A display is attachable to the bracket in one of the four bracket orientations. In an embodiment, the base is mounted on a top surface of a transaction terminal.

900



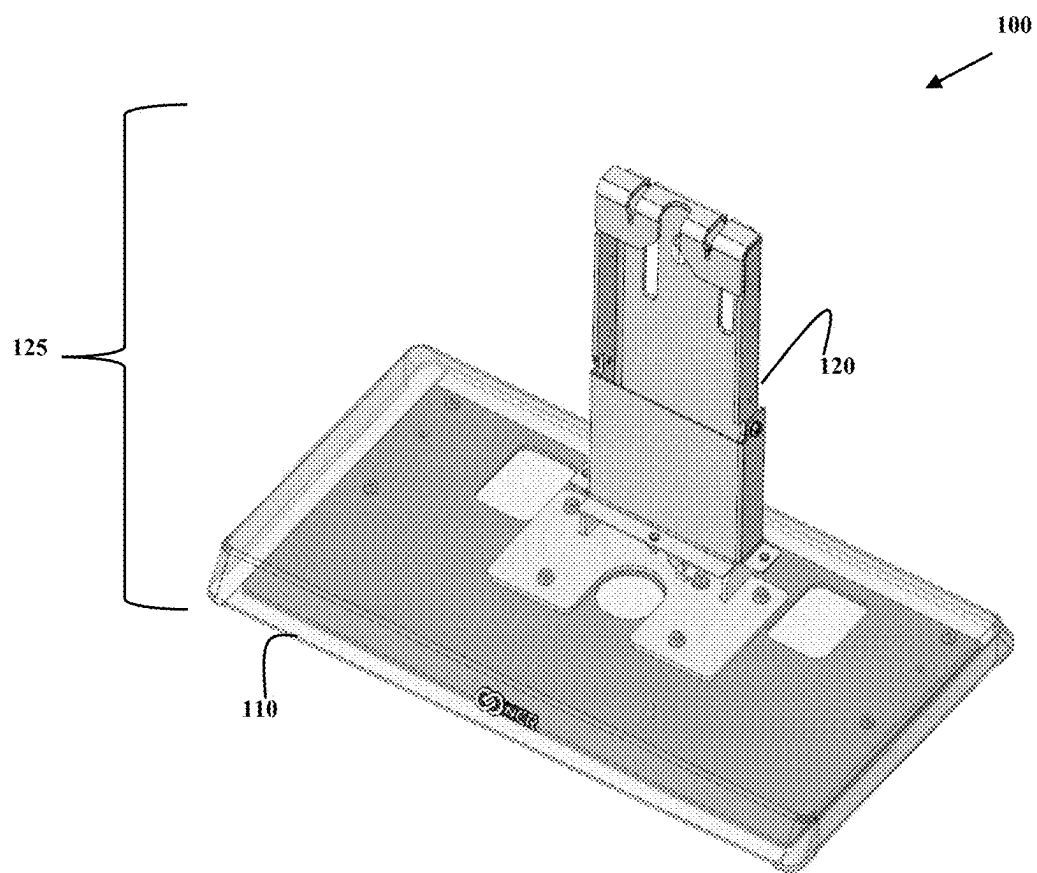


FIG. 1

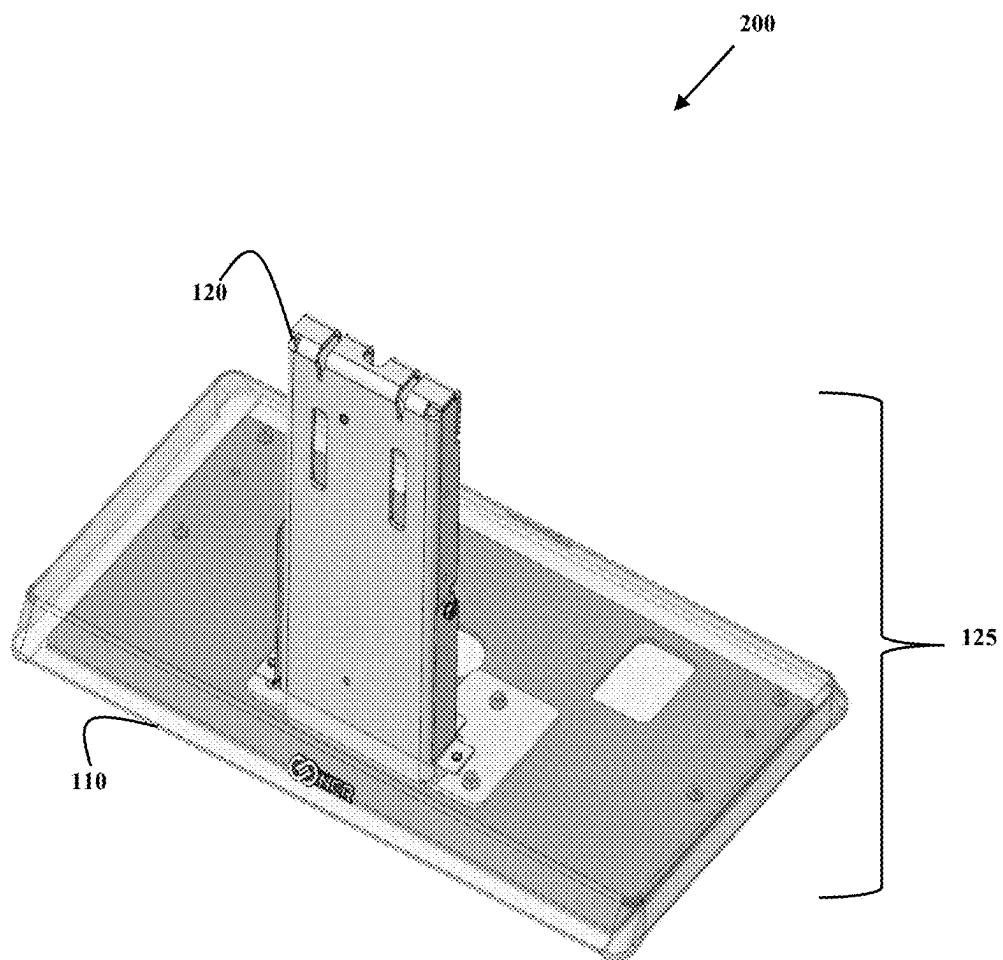


FIG. 2

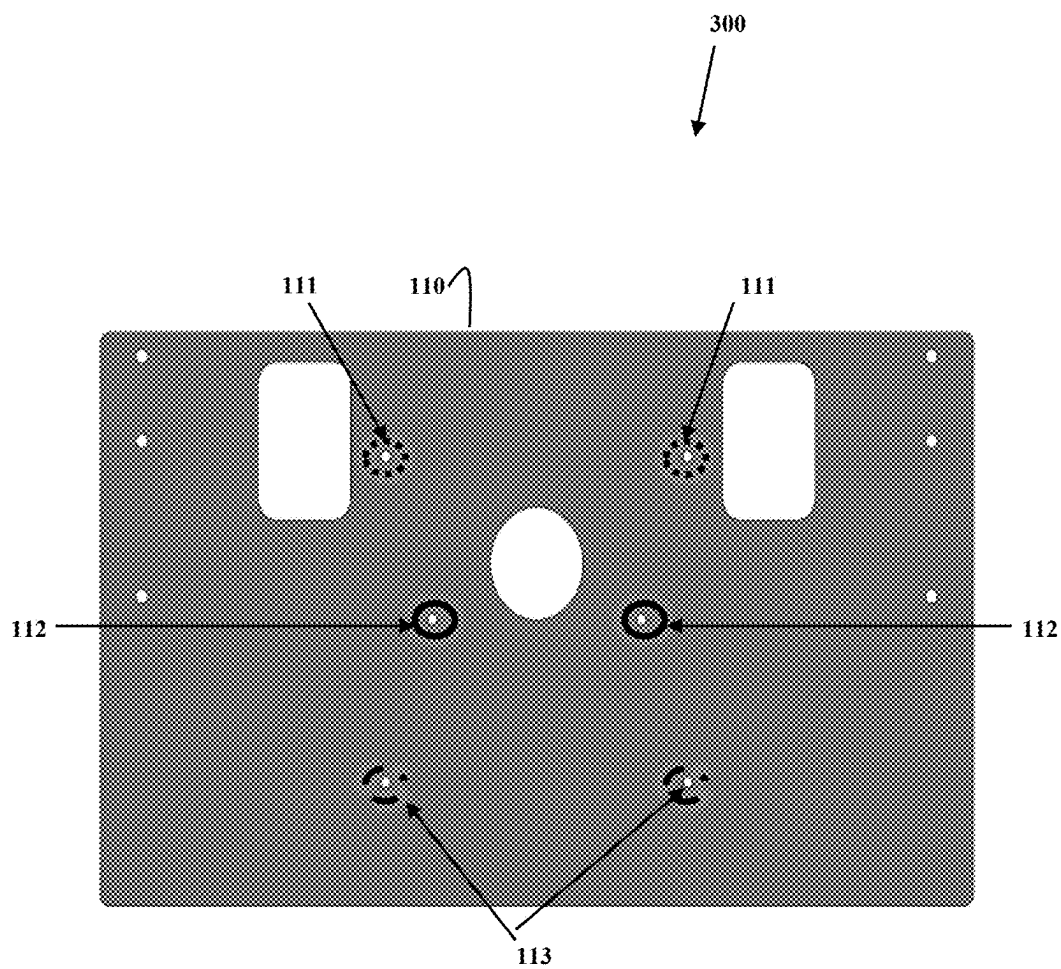


FIG. 3

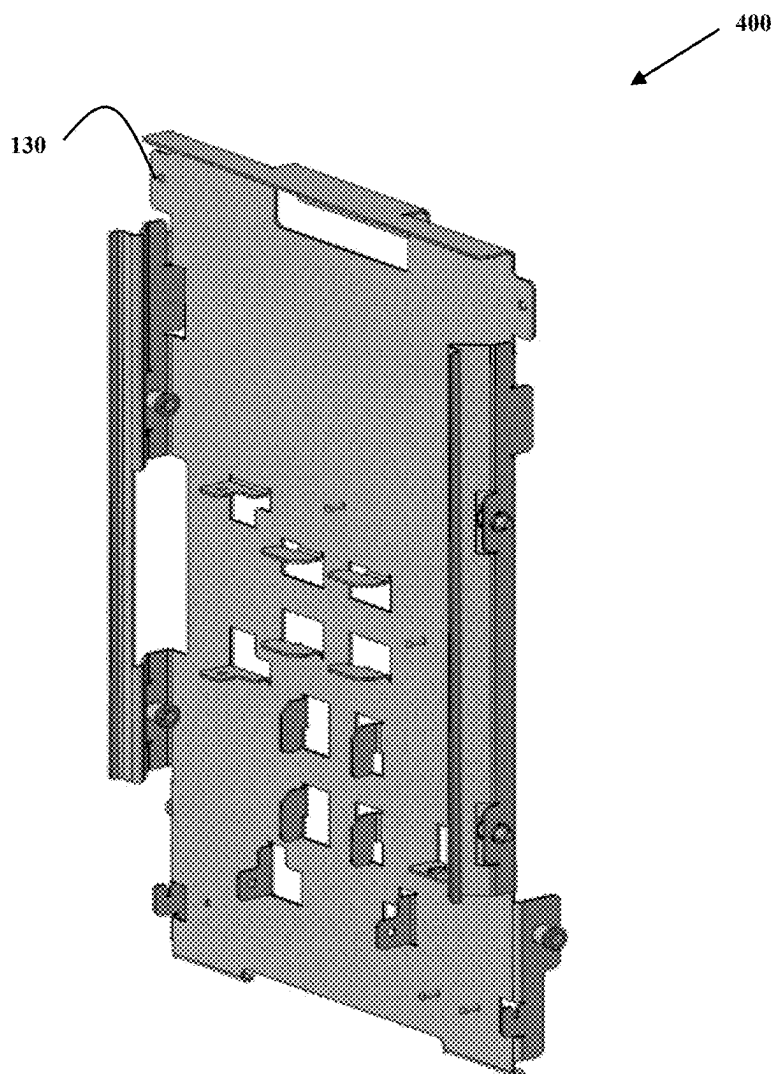
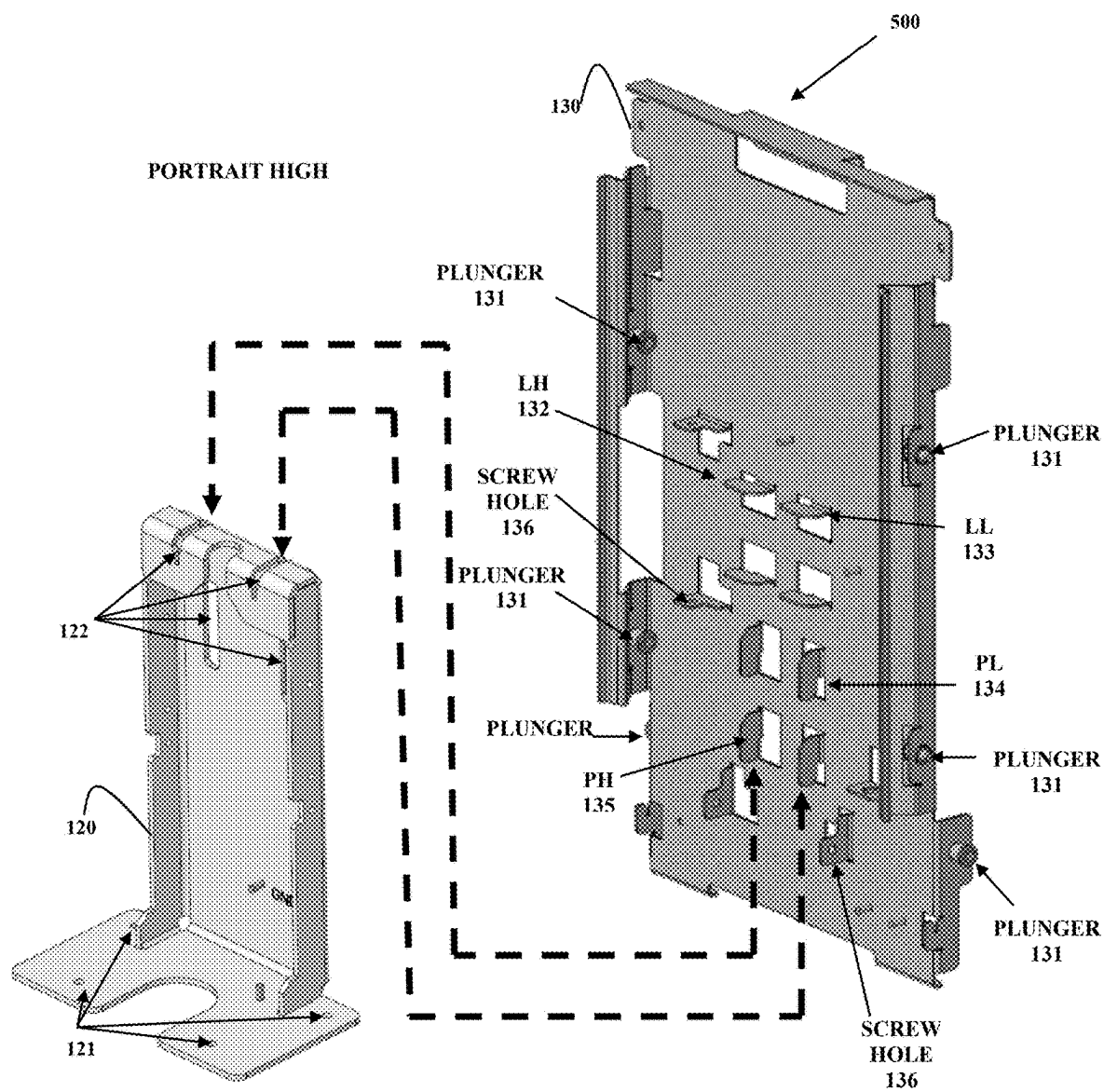


FIG. 4



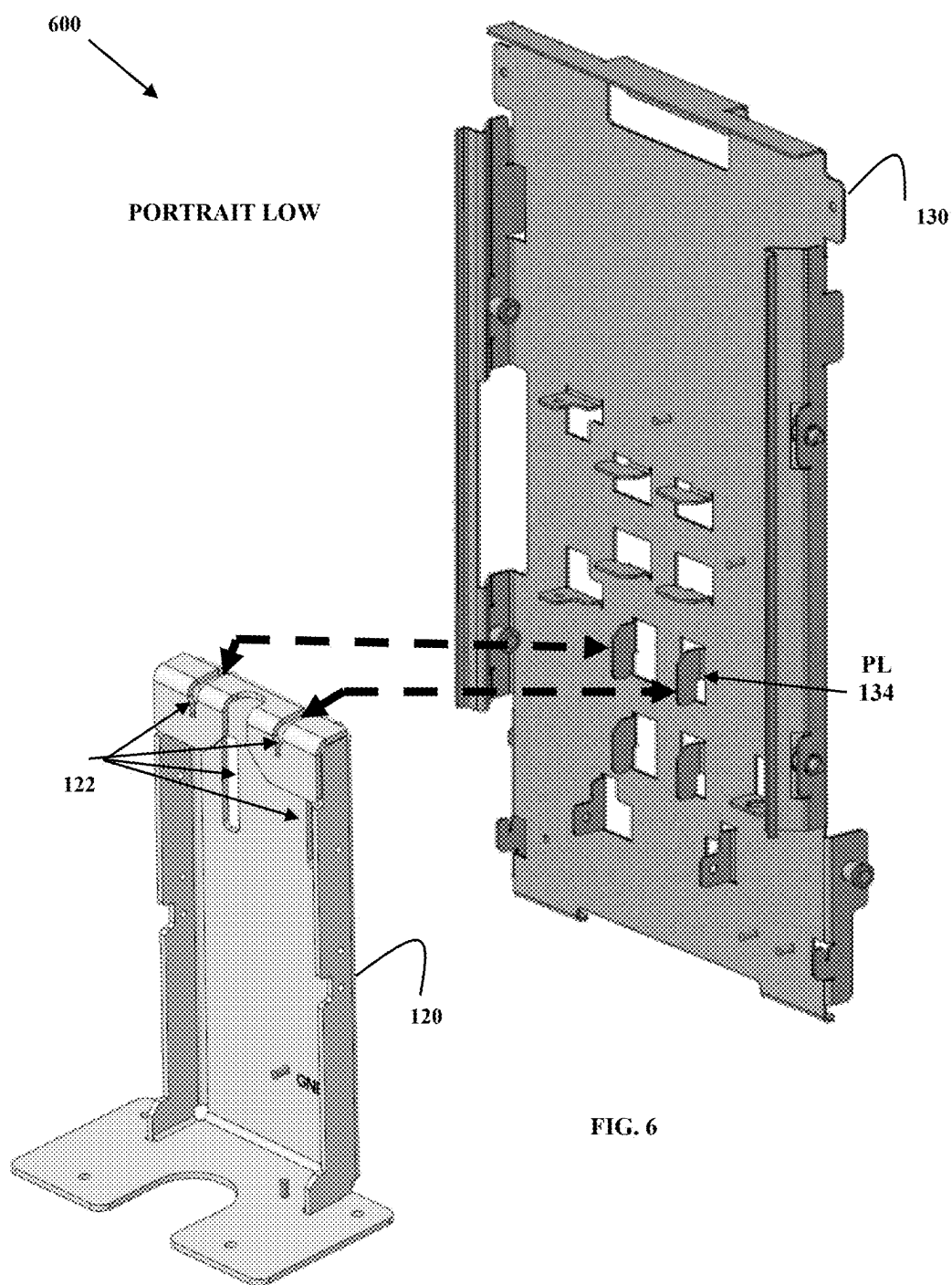
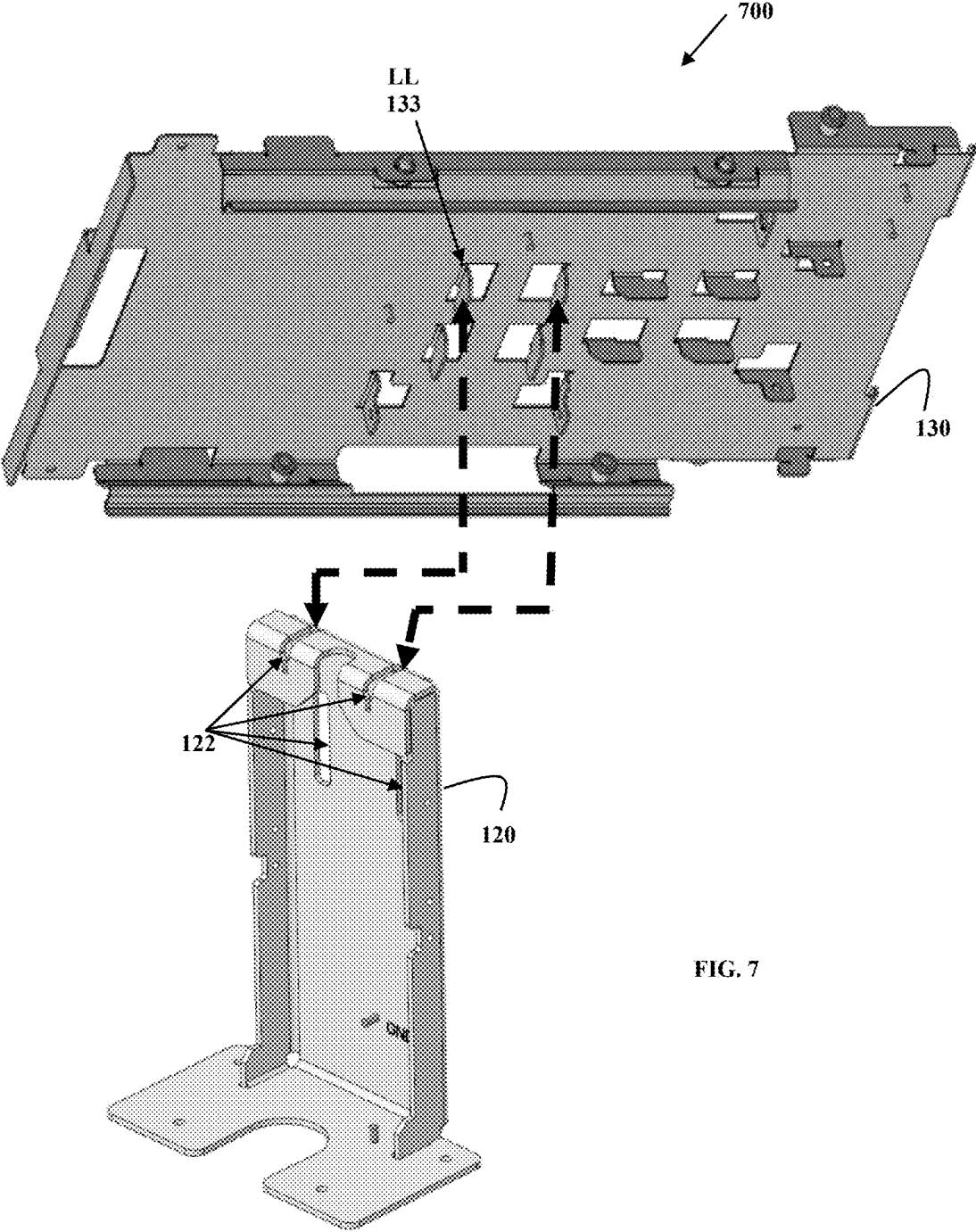


FIG. 6

LANDSCAPE LOW



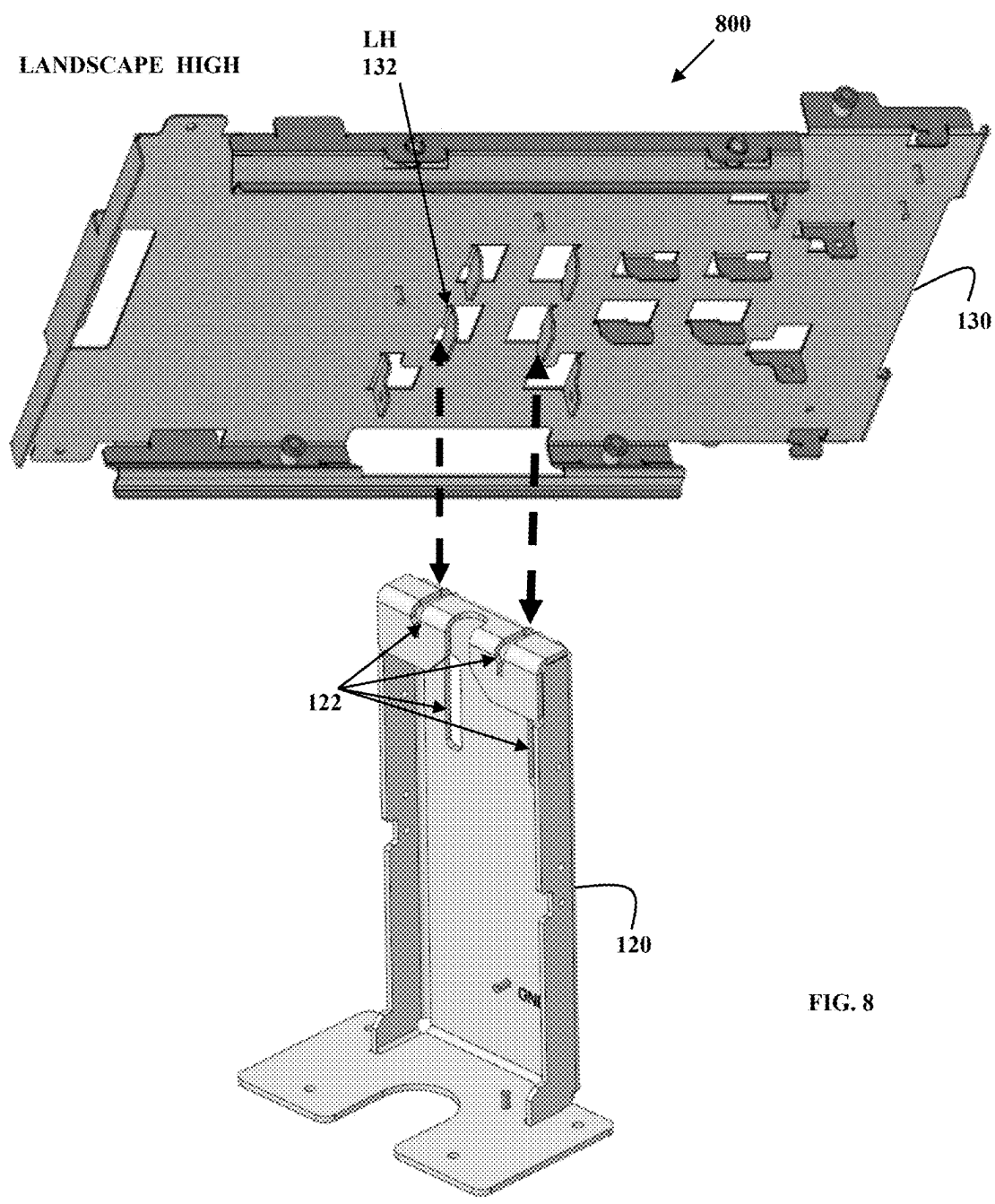


FIG. 8

900

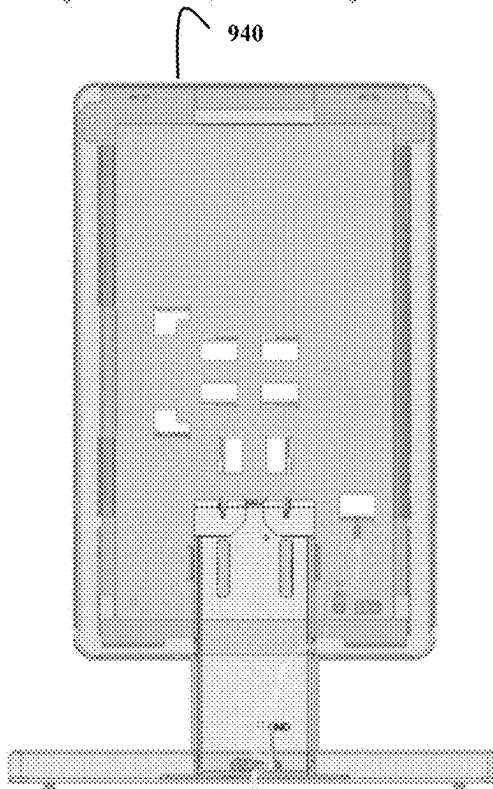
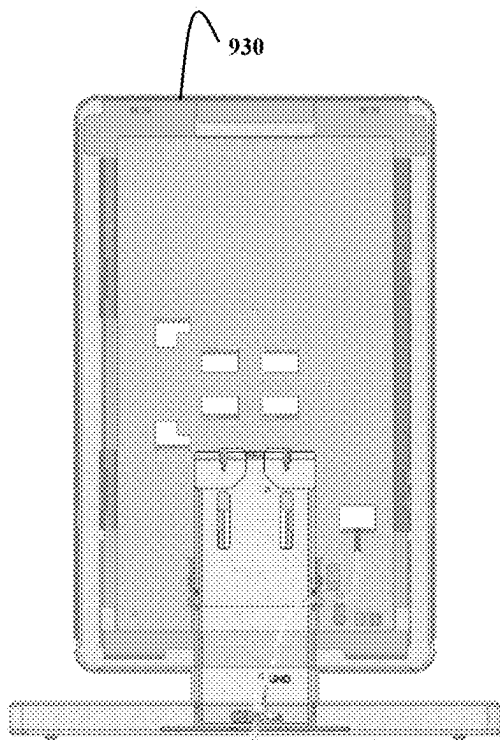
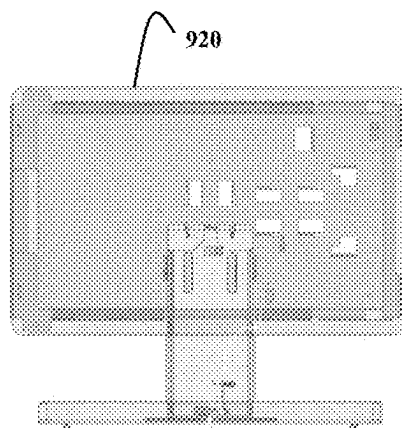
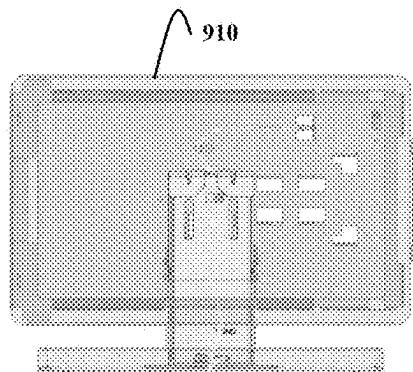


FIG. 9

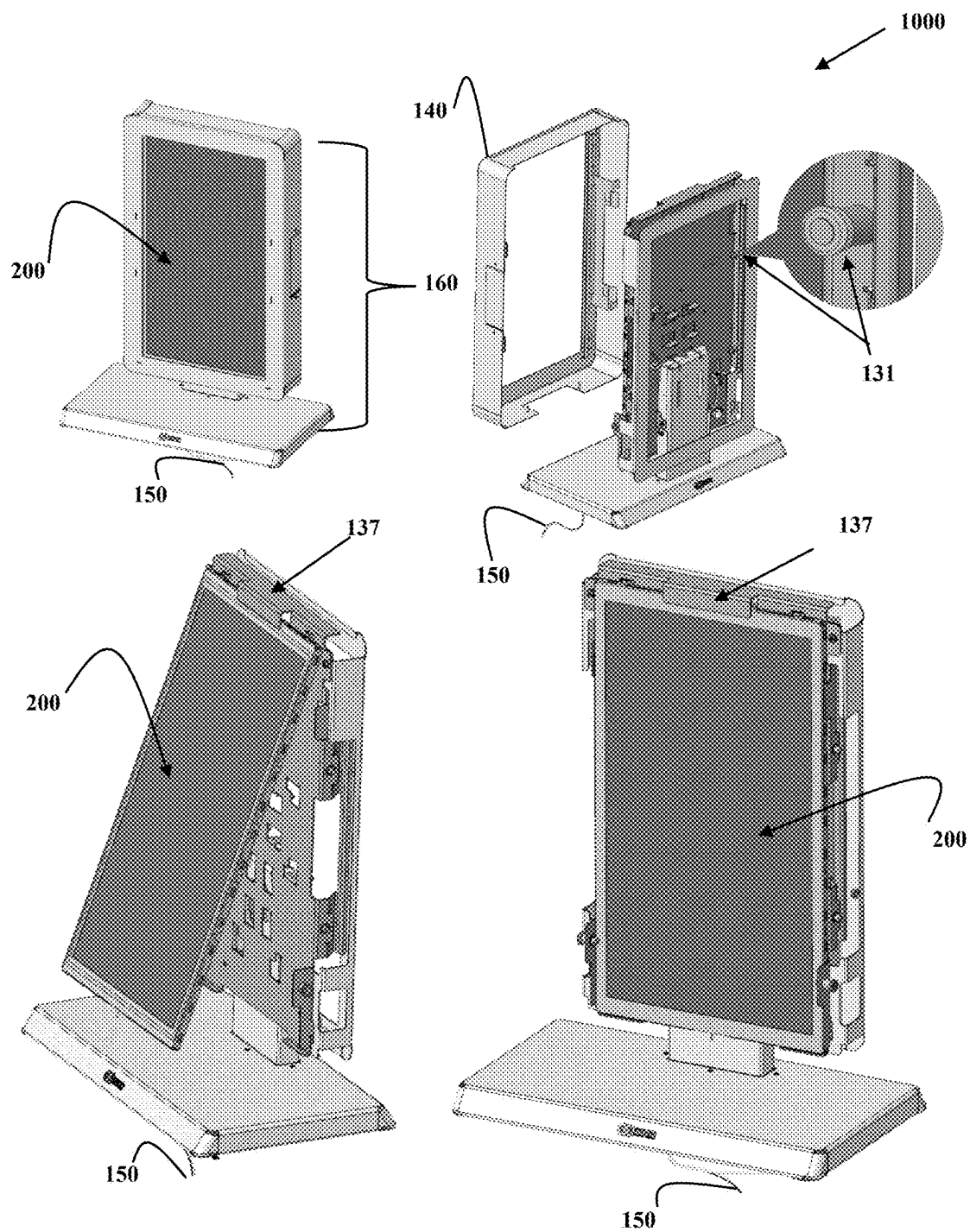


FIG. 10

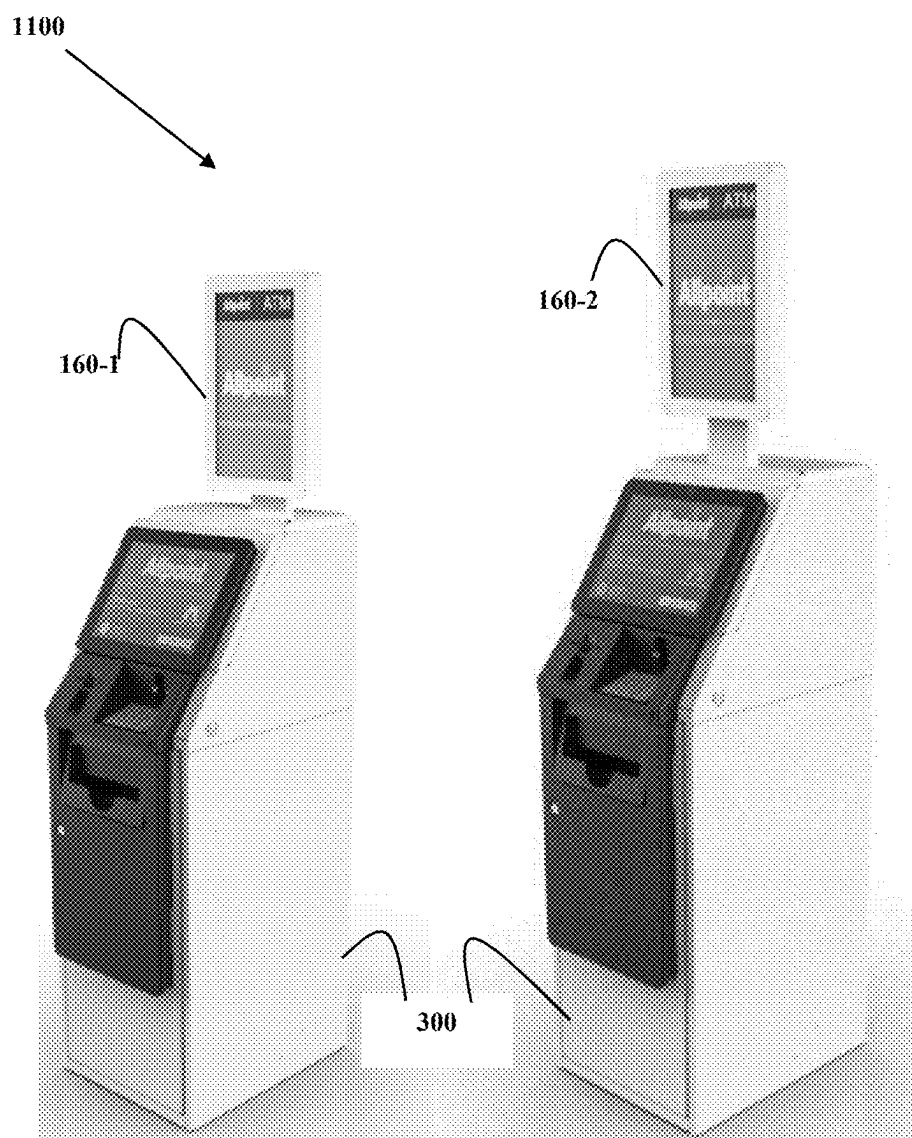


FIG. 11

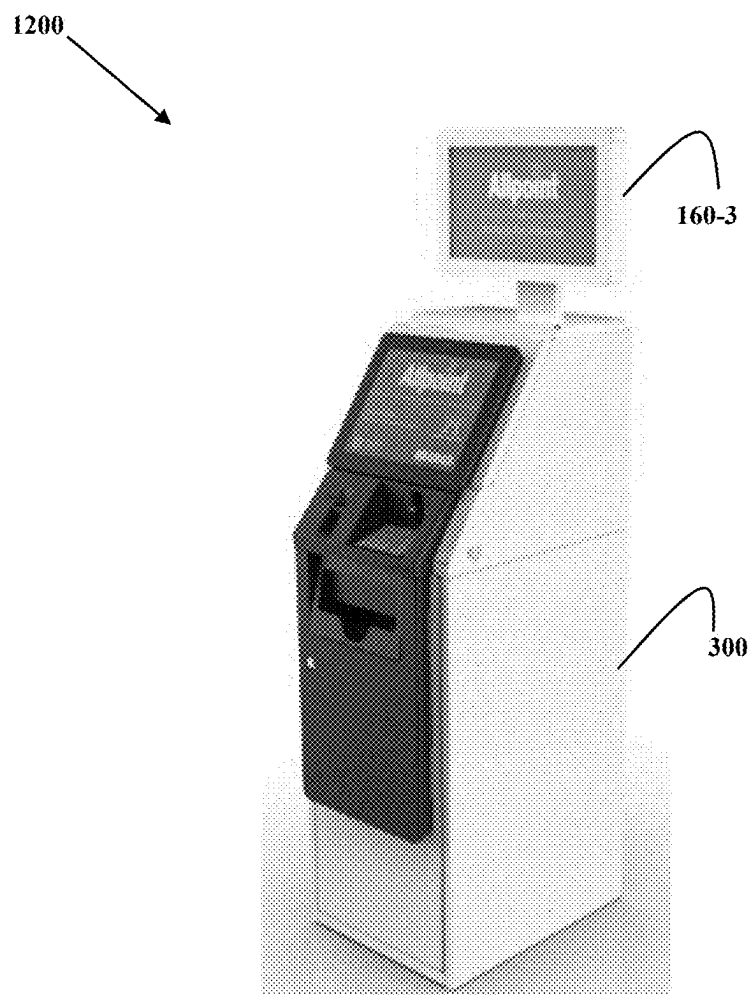


FIG. 12

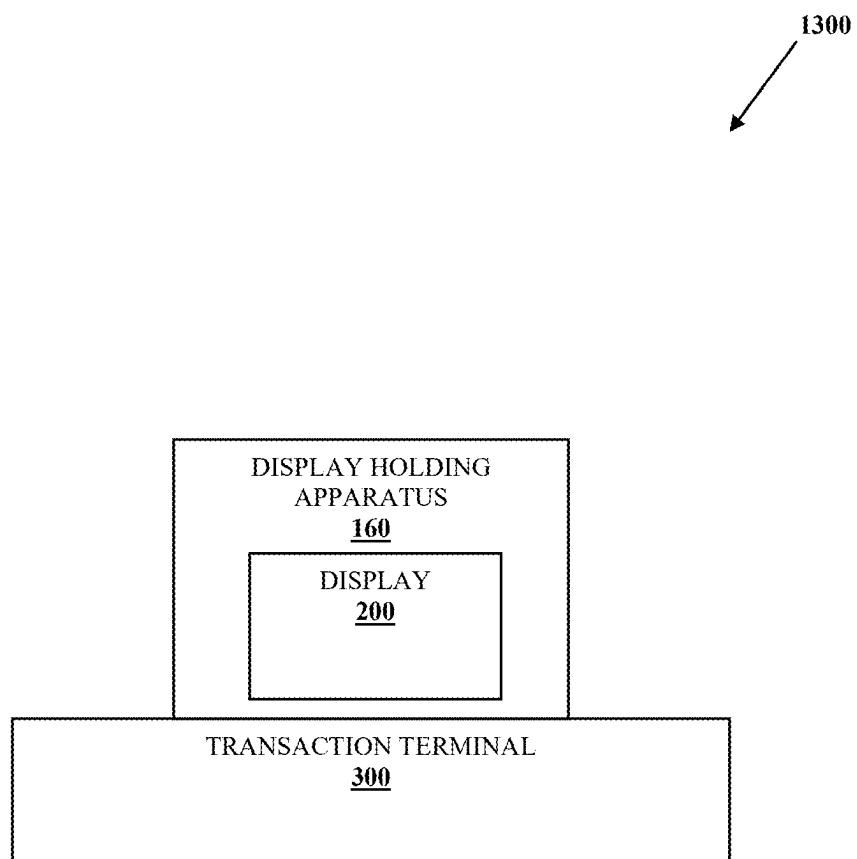


FIG. 13

CONFIGURABLE DISPLAY HOLDING APPARATUS AND SYSTEM

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a continuation of U.S. patent application Ser. No. 18/103,650, filed Jan. 31, 2023, which application and publication is incorporated herein by reference in its entirety.

BACKGROUND

[0002] Displays and/or illuminated signs come in all a variety of different shapes, sizes, and capabilities. Displays/signs are also popular for advertisements or for drawing a customer's attention to another device, such as an automated teller machine (ATM). Any a sign/display affixed to a top surface of an ATM is often referred to as a "topper."

[0003] Current designs of ATM toppers are bulky, heavy, and inflexible. There exists just a few design options and each option is restricted to a designated position and orientation on the top surface of the ATM. Existing toppers are also time consuming to assemble and are usually shipped to a retail site separately from the ATM.

SUMMARY

[0004] In various embodiments, a configurable display holding apparatus and system are presented. The apparatus includes a base, a vertical member, and a bracket. The vertical member is attached to the base to form a stand. The bracket attaches to the vertical member of the stand in a plurality of orientations. A display attaches to the bracket in the bracket's configured orientation with the vertical member of the stand.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a diagram of a stand for a display holding apparatus configured in a back orientation, according to an example embodiment.

[0006] FIG. 2 is a diagram of the stand for a display holding apparatus configured in a front orientation, according to an example embodiment.

[0007] FIG. 3 is a diagram of a base for the stand of the display holding apparatus, according to an example embodiment.

[0008] FIG. 4 is a diagram of a configurable bracket for attaching to the stand of the display holding apparatus, according to an example embodiment.

[0009] FIG. 5 is a diagram illustrating the attachment members of the configurable bracket to attach the bracket to the stand for a display in a high portrait orientation, according to an example embodiment.

[0010] FIG. 6 is a diagram illustrating the attachment members of the configurable bracket to attach the bracket to the stand for a display in a low portrait orientation, according to an example embodiment.

[0011] FIG. 7 is a diagram illustrating the attachment members of the configurable bracket to attach the bracket to the stand for a display in a low landscape orientation, according to an example embodiment.

[0012] FIG. 8 is a diagram illustrating the attachment members of the configurable bracket to attach the bracket to the stand for a display in a high landscape orientation, according to an example embodiment.

[0013] FIG. 9 are diagrams illustrating the bracket attached to the stand for four different display orientations, according to an example embodiment.

[0014] FIG. 10 are diagrams illustrating different sized displays held by the display holding apparatus, according to an example embodiment.

[0015] FIG. 11 are diagrams of transaction terminals with displays held by the display holding apparatus, according to an example embodiment.

[0016] FIG. 12 is another diagram of a transaction terminal with a display held by the display holding apparatus, according to an example embodiment.

[0017] FIG. 13 is a block diagram of a system that includes the display holding apparatus, according to an example embodiment.

DETAILED DESCRIPTION

[0018] As stated above, most transaction terminals, such as ATMs, include a topper affixed to a top surface of the ATMs for purposes of providing customer advertisements to customers passing by and/or for purposes of enticing customers to perform transactions on the ATMs. In most cases, these toppers are just lighted signs with fixed and rigid designs attached to the top surface of the ATMs. The dimensions, orientations, and position of the toppers on the ATM are not configurable. More often than not, the retailer selects a topper design and then has a sign panel custom made in the size and dimensions supported by the topper design; a light behind or in the panel provides the retailer's advertisement. Consequently, many toppers appear out of place with the overall aesthetic design of the ATM or the surroundings of where the ATM is located.

[0019] Customers frequent transaction terminals to perform transactions, such as to purchase an item, withdraw currency, deposit currency, place an order with a retailer, review a status of an order, make a reservation, check-in to an establishment, etc. Accordingly, the attention of a customer is captured during a transaction at a terminal. This makes for prime real estate for retailers to pitch their products and services or to establish a retailer's brand with the customer. Additionally, displays of the terminals are often intentionally obfuscated to passers-by so that a customer performing a transaction has privacy; thus, the space above the terminal display can also be used to attract a customer to the terminal for a transaction.

[0020] The teachings provided address these issues by providing a configurable display holder apparatus and system. The apparatus can be configured to hold a lighted panel display or an intelligent display in eight different configurations, four for portrait display orientation and four for landscape display orientation. The apparatus includes a minimal amount of components and can be assembled in minutes. Moreover, the apparatus can be reconfigured for a different display orientation from an existing display orientation in minutes. The components of the apparatus are less than existing components of existing industry toppers; weigh less resulting in reduced shipping expenses; and can be setup and/or reconfigured without any skilled technician in minutes.

[0021] As used herein a "topper" is a device that includes panels which are lighted to produce a visual branding or advertisement on top of a transaction terminal. A "transaction terminal" is a self-service terminal (SST), an automated teller machine (ATM), or a kiosk. A "display" can include a

topper with lighted panels to visually convey a message or a branding, a monitor, a digital sign, or an independent processing device integrated into the monitor (e.g., a tablet, touch-based monitor, smart/intelligent monitor/display, etc.).

[0022] The display apparatus 160 (shown in FIG. 10 below) includes a stand 125 (shown in FIG. 1 below), a bracket 130, a bracket and display cover 140 (shown in FIG. 10 below), and optionally a display 200 (shown in FIG. 10 below). FIG. 1 is a diagram 100 of a stand 125 of a display holding apparatus 160 configured in a back orientation, according to an example embodiment. FIG. 2 is a diagram 200 of the stand 125 of a display holding apparatus 160 configured in a front orientation, according to an example embodiment. FIG. 3 is a diagram 300 of a base 110 for the stand 125 of the display holding apparatus 160, according to an example embodiment.

[0023] The base 110 includes six premanufactured screw holes or apertures 111, 112, and 113. The back orientation shown in FIG. 1 utilizes screw holes 111 and 112 for purposes of securing a vertical member 120 to the base 110 to form the stand 125. The front orientation shown in FIG. 2 utilizes screw holes 112 and 113 for purposes of securing vertical member 120 to base 110 to form the stand 135. The vertical member 120 attaches to the base 110 at a right angle to the base 110. Screw holes 112 are common to both the back and front orientations of the stand 125. Corresponding screw holes 121 are shown at the bottom of vertical member 120 in FIG. 5 below.

[0024] The four screw holes 121 of vertical member 120 only align with four of the six screw holes 111, 112, and 113. This ensures that during setup the vertical member 120 can be easily aligned with base 110 and mounted via 4 screws as stand 125 to a top surface of a terminal. The four aligned screw holes 121 form a back orientation when aligned to 111 and 112 of base 110. A back orientation is one in which the vertical member 120 sits farther away from the front-edge of the terminal. The four aligned screw holes 121 form a front orientation when aligned to 112 and 113 of base 110. The front orientation is one in which the vertical member 120 sits closer to the front-edge of the terminal.

[0025] FIG. 4 is a diagram 400 of a configurable bracket 130 for attaching to the stand of 125 the display holding apparatus 160, according to an example embodiment. Bracket 130 includes a plurality of manufactured tabs and screw holes for affixing the bracket 130 to stand 120 and movable plugs for affixing a display 200 (shown in FIG. 10 below). Bracket 130 can be configured to orient an attached display in four different orientations: a low portrait orientation (shown in FIGS. 6 and 9 below), a high portrait orientation (shown in FIGS. 5 and 9 below), a low landscape orientation (shown in FIGS. 7 and 9 below), and a high landscape orientation (shown in FIGS. 8 and 9 below). The bracket 130 adapted to attach to the vertical member 120 in parallel with the orientation of the vertical member 120 or at a right angle to the vertical member 120. The in parallel attachment of the bracket 130 relative to the vertical member 120 is shown in FIGS. 5 and 6 and the bottom two diagrams in FIG. 9. The right-angle attachment of the bracket 130 relative to the vertical member 120 is shown in FIGS. 7 and 8 and the top two diagrams in FIG. 9.

[0026] FIG. 5 is a diagram 500 illustrating the tab members 135 of the configurable bracket 130 to attach the bracket to the vertical member 120 of the stand 125 for a display 200

in a high portrait orientation, according to an example embodiment. Vertical member 120 includes manufactured slots 122, which tabs 135 slide of bracket 130 slide into to configure bracket 130 for holding a display 200 in a high portrait orientation.

[0027] Tabs 134 are inserted into slots 122 to configure bracket 130 for holding a display 100 in a low portrait orientation. Bracket 130 can also be rotated 90 degrees relative to vertical member 120 and tabs 132 and 133 inserted into slots 122 for holding a display in high or low landscape orientation. Plungers 131 pull out and include a protective rubber tip for inserting edges a display 200 and/or for inserting a back surface of display 200 against bracket 130. When the plungers 131 are released, they push into the sides/back surface of display 200 and hold display 200 firmly against bracket 130.

[0028] FIG. 6 is a diagram 600 illustrating the tab members of the configurable bracket 130 to attach the bracket 130 to the stand 125 for a display 200 in a low portrait orientation, according to an example embodiment. Tabs 134 are inserted into slots 122 with tabs 135 beneath tabs 134. Tabs 135 fit through other slots 122 such that the bottom of tabs 134 rest on top of vertical member 120 in top slots 122. This lowers bracket 130 below a vertical height of bracket 130 from that which was shown in FIG. 5 for the high portrait orientation. In the high portrait orientation as shown in FIG. 5 tabs 134 are above tabs 135 which rest on a top of vertical member 120 elevating bracket 130 to a higher vertical position than what is shown in FIG. 6.

[0029] FIG. 7 is a diagram 700 illustrating the attachment tab members of the configurable bracket 130 to attach the bracket 130 to the stand 125 for a display 200 in a low landscape orientation, according to an example embodiment. Bracket 130 is rotated 90 degrees counter clockwise relative to vertical member 120. Tabs 133 are inserted into top slots 122 of vertical member 120 to configure a low landscape orientation for any subsequently attached display 200.

[0030] FIG. 8 is a diagram 800 illustrating the tab members of the configurable bracket 130 to attach the bracket 130 to the stand 125 for a display 200 in a high portrait orientation, according to an example embodiment. Once again bracket 130 is rotated 90 degrees counter clockwise relative to vertical member 120. Tabs 132 are inserted into top slots 122 of vertical member 120 and rests on top of vertical member 120. Notice tabs 133 sit vertically above tabs 132 and are not inserted into any tabs in vertical member 120 this provides a higher vertical position for bracket 130 relative to the vertical member 120 than what was shown in FIG. 7 for the low landscape orientation. When tabs 132 are inserted into top slots 122 of vertical member 120 any subsequently attached display 200 is in a high landscape orientation.

[0031] FIG. 9 are diagrams 800 illustrating the bracket 130 attached to the stand 125 on vertical member 120 for four different display orientations, according to an example embodiment. The top rightmost figure shows a low landscape orientation of display apparatus 160 and is configured in the manner discussed above with FIG. 7. The top leftmost figure shows a high landscape orientation of display apparatus 160 and is configured in the manner discussed above with FIG. 8. The bottom rightmost figure shows a low portrait orientation of display apparatus 160 and is configured in the manner discussed above with FIG. 6. The bottom

leftmost figure shows a high portrait orientation of display apparatus **160** and is configured in the manner discussed above with FIG. 5.

[0032] FIG. 10 are diagrams **1000** illustrating different sized displays **200** held by the display holding apparatus **160**, according to an example embodiment. An optional display bezel or cover **140** is shown that fits around a periphery of the display **200** attached to bracket **130** of display apparatus **160**. Diagrams **1000** also illustrate a top tab **137** of bracket **130** under which the display **200** is tucked under. Plungers **131** are pulled out and released causing plungers **131** to push against and hold the sides of display **200** against bracket **130**. Cover **140** can optionally be placed around the periphery of the display **200**. The diagrams **1000** of FIG. 10 illustrate the display apparatus **160** configured in a back and low portrait orientation.

[0033] A desired display **200** is attached to display holding apparatus **160** with a power and/or power and data cord **150** from lights, light emitting diodes (LEDs) board, and/or data port of display **200** running underneath and through base **110** of display holding apparatus **160**. This allows the power or combined power and data wire **150** to be connected accordingly to the terminal without any visible cord **150** when a bottom side of base **110** is mounted to a top side surface of the terminal.

[0034] It is noted that for each of the four discussed display apparatus orientations. The display apparatus **160** can also be in a front orientation (i.e. front edge of stand **120** is closer to the front edge of the terminal) as shown in FIG. 2 or display apparatus **160** can be back orientation (e.g., front edge of stand **120** is farther from the front edge of the terminal) as shown in FIG. 1. Thus, there are 8 possible configurations achievable with display apparatus **160** via just three components for base **110**, stand **120**, and bracket **130** or via four components for optional display cover **140**.

[0035] In an embodiment, four of the six screw holes **111**, **112**, and **113** in base **110** are manufactured to align with existing screw receptacles (**4**) in a top surface of a transaction terminal. Screw holes **112** align with at least two of the four existing screw receptacles and whatever orientation front or back selected by a customer for the display holding apparatus **160** ensures that at least two of screw holes **111** or two of screw holes **113** align with the remaining two existing screw receptacles. Thus, apparatus **160** can be used to replace an existing topper of a terminal with just 3 screws and a desired configuration of eight potential configurations setup without any screws or any tools being required.

[0036] In an embodiment, the display holding apparatus **120** includes an LED board and a panel that sits in front of the LED board to form a display **200** through control of the LEDs on the board illuminated through the panel. In an embodiment, display holding apparatus is sized to hold a tablet touch computer display **200**.

[0037] In an embodiment, bracket **130** is resizable to hold different sized displays **200**. For example a central rail running horizontally across the bracket **130** and a central rail or pair of side rails running vertically across the bracket can be expand out to increase the size of the bracket horizontally and vertically or fold/tuck in to decrease the size of the bracket horizontally and vertically. In this way, display holding apparatus **160** can be custom sized to accommodate different sized displays **200**.

[0038] FIG. 11 are diagrams **1100** of transaction terminals **300** with displays held by the display holding apparatus **160**, according to an example embodiment. The leftmost figure shows the display apparatus **160-1** configured in a low portrait orientation for the display **200** and a back orientation for stand **120**. The rightmost figure shows the display apparatus **160-2** configured in a high portrait orientation for the display **200** and a front orientation for stand **120**. Although not shown, there can also be a low portrait orientation in a front orientation for the stand and a high portrait orientation in a back orientation from stand **120**.

[0039] FIG. 12 is another diagram **1200** of a transaction terminal **300** with a display **200** held by the display holding apparatus **160**, according to an example embodiment. The display **200** is in a high landscape orientation and a back orientation for the stand **120** on top of terminal **300**. Although not shown, there can also be a high landscape orientation for display **200** in a front orientation for the stand **120**, a low landscape orientation for display **200** in a front orientation for the stand **120**, and a low landscape orientation for display **200** in a back orientation for the stand **120**. [0040] Again, there can be any of four different display orientations and each of the four can also be in either a front orientation for the stand **120** or a back orientation for the stand **120**. Thus, there are eight different configurations achievable with display apparatus **160**.

[0041] FIG. 13 is a block diagram of a system **1300** that includes the display holding apparatus **160**, according to an example embodiment. It is noted that system **1300** is shown in greatly simplified form with only those components necessary for understanding the embodiments provided. Thus, there can be more or less components without departing from the teachings provided herein.

[0042] System **1300** includes a transaction terminal **300**, display holding apparatus **160**, and display **200**. The display holding apparatus **160** includes a base **110**, a vertical member **120**, and a bracket **130**. In an embodiment, display holding apparatus **160** includes a display cover **140**. In an embodiment, display holding apparatus **160** includes an LED light board and a light panel. In an embodiment, the display **200** is a lighted sign with lights and one or more covers/panels, a monitor, a touch monitor, or a tablet.

[0043] The vertical member **120** is attachable to base **110** in a front orientation or a back orientation. The bracket **130** is attachable to the vertical member **120** in a high landscape orientation, a low landscape orientation, a high portrait orientation, or a low portrait orientation. Thus, display holding apparatus **160** can be configured in 8 different configurations.

[0044] In an embodiment, the apparatus **160** when holding a specific display **200** forms a display system.

[0045] The above description is illustrative, and not restrictive. Many other embodiments will be apparent to those of skill in the art upon reviewing the above description. The scope of embodiments should therefore be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

[0046] In the foregoing description of the embodiments, various features are grouped together in a single embodiment for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting that the claimed embodiments have more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all

features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Description of the Embodiments, with each claim standing on its own as a separate exemplary embodiment.

1. (canceled)
2. A method, comprising:
 - receiving an electromagnetic noise signal on a first read head of a card reader;
 - receiving the electromagnetic noise signal on a second read head of the card reader;
 - transforming electromagnetic waves associated with the electromagnetic noise signal on the second read head and applying to the first read head to cancel the electromagnetic noise signal on the first read head; and
 - reading magnetic encoded information on a magnetic strip of a card inserted into a card slot of the card reader while preventing any skimmer from reading the magnetic encoded information on the magnetic strip because of a presence of the electromagnetic noise signal.
3. The method of claim 2, wherein transforming further includes connecting the second read head to the first read head in antiphase to achieve the transforming.
4. The method of claim 2, wherein receiving the electromagnetic noise signal on the first read head further includes receiving the electromagnetic noise signal from a disruptive electromagnetic circuit of the card reader.
5. The method of claim 2, wherein receiving the electromagnetic noise signal on the first read head further includes receiving the electromagnetic noise signal from a disruptive electromagnetic source generator that is external to the card reader.
6. The method of claim 2, wherein reading further includes aligning the first read head within the card slot to be situated under track 2 of the magnetic strip of the card.
7. The method of claim 2, wherein reading further includes elevating the first read head at a height above a top of the second read head.
8. The method of claim 7, wherein elevating further includes setting the height to approximately 1 to 2 millimeters.
9. The method of claim 2, wherein reading further includes connecting a first terminal and a second terminal together through the first read head.
10. The method of claim 9, further comprising connecting a third terminal and a fourth terminal together through the second read head.
11. The method of claim 10, further comprising connecting the third terminal to the first terminal and connecting the fourth terminal to the second terminal.
12. The method of claim 2, wherein reading further includes reading the magnetic encoded information through a motorized card reader as the card reader.

13. A method, comprising:
 - receiving a card within a card slot of a terminal;
 - activating an electromagnetic noise signal within the terminal;
 - providing the electromagnetic noise signal to a card reader integrated into the terminal;
 - receiving magnetic encoded information from the card reader after the card reader processes the electromagnetic noise signal to cancel effects of the electromagnetic noise signal on a first read head while maintaining the electromagnetic noise signal on a second read head; and
 - processing a transaction using the magnetic encoded information.
14. The method of claim 13, wherein activating further includes generating the electromagnetic noise signal from a disruptive electromagnetic source generator of the terminal.
15. The method of claim 13, wherein providing further includes sending the electromagnetic noise signal to both the first read head and the second read head of the card reader.
16. The method of claim 13, wherein receiving further includes obtaining the magnetic encoded information from track 2 of a magnetic strip of the card.
17. The method of claim 13, wherein processing further includes processing the transaction as an automated teller machine (ATM) transaction.
18. The method of claim 13, wherein processing further includes processing the transaction as a point-of-sale (POS) transaction.
19. The method of claim 13, wherein processing includes processing the transaction as a self-service terminal (SST) transaction.
20. A system, comprising:
 - a card reader comprising:
 - a first read head;
 - a second read head connected to the first read head;
 - wherein the first read head and second read head are configured to:
 - receive an electromagnetic noise signal,
 - transform electromagnetic waves associated with the electromagnetic noise signal received on the second read head,
 - apply the transformed electromagnetic waves to the first read head to cancel the electromagnetic noise signal on the first read head, and
 - permit the first read head to read a magnetic strip of a card while the electromagnetic noise signal remains active on any skimmer present within the card reader.
21. The system of claim 20, wherein the first read head is elevated at a height above the second read head and aligned under track 2 of the magnetic strip when the card is inserted into a card slot of the card reader.

* * * * *